Yafira Martinez

Design Engineer / Creative Technologist

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Interactive Systems, Interfaces & Experimental Technologies

SKILLS

Front-End & Web Development: React, Next.js, TypeScript, JavaScript, HTML/CSS, Node.js, Astro, REST APIs, Web Serial, p5.js, ml5.js, Creative Coding, CMS, Databases (Supabase, Redis, SQL), Vercel, Git/GitHub

Design: Interaction Design, UX/UI, Responsive Web Design, Design Systems, Design Tokens, Prototyping, Wireframing, Digital Accessibility, Information Architecture, User Research, Figma, Adobe Suite (Illustrator, Photoshop, XD)

Hardware: Physical Computing, Microcontrollers (Arduino, Raspberry Pi, ESP32, RP2040), CircuitPython, Python, E-Ink & TFT Displays, Capacitive Sensing, Haptics, E-Textiles, Soft Circuits, Wearables, Soldering, Fabrication (laser cutting, 3D printing, knitting machine)

EXPERIENCE

Creative Technologist, Brand Systems — Kurppa Hosk Oct 2026 – Present
Contract

- Explore brand intelligence through creative sprints for brand and communications; evaluate the studio's Brand OS prototype.
- Research emerging creative tools and share findings and recommendations with the brand team.

Founder, Design Engineer & Creative Technologist — Electrocute Lab Jan 2023 – Present
Independent creative practice

- Design and ship web tools end to end, including electrocute-ui, a React component library with an 18-color design token system published to npm, and tinytinker.tools, an open-source maker toolkit of 22+ browser tools.
- Build soft circuits, e-textiles, physical computing, and generative text work exhibited internationally, alongside open workshop materials and a technical blog (Astro + TinaCMS).

Lead Camp Counselor — ITP Camp, New York University May 2026 – Jul 2026
Previously: Camp Counselor, Jun – Jul 2025

- Led a 150+ person creative technology community, mentoring first-year counselors and supporting hardware, software, and fabrication.
- Built spot-mapper, a real-time collaborative floor-plan mapping tool (Leaflet, Vercel, Redis) used live at the Show All Things Show.

Machine Learning Researcher & Web Contributor — ml5.js Jun 2025 – May 2026
Mentors: Daniel Shiffman & J.H. Moon

- Researched and prototyped ml5.transcribe() and ml5.embeddings(), bringing browser-based transformer models (Whisper, Moonshine) to a library built for students, artists, and beginner coders.
- Built two Docsify plugins for the docs site (inline code explanations, copy-to-clipboard), featured on the ml5.js blog.

Hardware Research Developer — NYU Ability Lab Oct 2025 – May 2026

- Researched accessible fashion technologies for Making Fashion Accessible, using knitting machines and AYAB (open-source knitting software) to support inclusive, adaptable garment production.

Lead Web Developer — Equitable Syllabus Project, NYU ITP/IMA Nov 2024 – May 2026

- Redesigned and rebuilt the project website with accessible, user-centered interfaces, from Figma designs to a production Next.js deployment, while migrating project data.

Web Developer & Designer — Useless Machines Archive, NYU Mar 2025 – May 2026
Professor: Blair Simmons

- Designed and built a front-end web archive using the Are.na API, with client-side data fetching and responsive interface design.

Software Engineer & Creative Technologist Intern — Floreo Labs Jan 2020 – May 2020

- Developed responsive React front-ends and designed UI/UX prototypes in Adobe XD within agile workflows.

EDUCATION

MPS, Interactive Telecommunications Program — New York University, Tisch School of the Arts May 2026
BS, Computer Science — City University of New York, Lehman College Dec 2020
Experimental Programming (Creative Coding & Design) — School for Poetic Computation (remote) 2022 – 2025
BBA, Business Administration (International Business) — City University of New York, Lehman College Sep 2012

Certifications — AI, Samsung Innovation Campus (2023) · HCI, Interaction Design Foundation (2022) · Google UX Design (2021)

SELECTED PROJECTS

The Soft Computer — Textile computing interface of machine-knit wool, capacitive fabric inputs, haptics, and a flexible e-ink display; a Raspberry Pi 5 drives Markov chain text across four corpus-mapped channels. Its second version, **The Soft Computer Mini**, is a portable iteration that composes one-of-a-kind e-ink tarot cards.

PetalByte — Portable generative text device treating the USB drive as archive and interface, housed in a flower-shaped laser-cut acrylic enclosure with a custom pygame interface.

Pocket Oracle of Time — Poetic timekeeping object exploring ritual, slowness, and material interaction with time through embedded electronics.

computer.garden() — Generative sketch that grows soft-circuit gardens using BFS maze-routing, rendered in a Bayer-dithered e-ink aesthetic.

EXHIBITIONS

They Make You Write Your Poems — Onassis ONX, NYC
The Soft Computer, Pocket Oracle of Time, Ribbon Logic · curated by Giulia Colletti
Oct 2026

Processing Community Day @ NYC
computer.garden()
Oct 2026

Bring Your Own Data — NY Tech Week (a16z), NYC
PetalByte
Jun 2026

Open Hardware Summit — Berlin, Germany
The Soft Computer Mini
May 2026

Periphery: ITP Thesis Show — NYC Resistor
The Soft Computer
May 2026

Electronics Faire — Temple University Libraries
Soft Circuit Library
Apr 2026

Complications — Horological Society of New York
Pocket Oracle of Time
Feb – Apr 2026

Pedestal Playtime — NYC Resistor
Moon Pocket
Jan 2026

ITP NIME Performance — Littlefield, NYC
Tendertronic (live analog circuit performance)
Dec 2025

Volt Couture — NYC Resistor
Static Chic
May 2025

Radiant Fibers — Clive Davis Museum, Brooklyn, NY
Soft Glitch (e-textile swatch)
Jan – Mar 2025

HACKATHONS

PlayfulNYC — Best of Software Apr 2026

Hard Mode: Hardware × AI, MIT Media Lab Mar 2026

UN Open Source Week Hackathon Jun 2025

MIT Reality Hack Jan 2025

TEACHING & WORKSHOPS

Tiny Soft Computers — SOFTER Digital Futures
Denmark · co-led with Alanna Okun · 16 participants
Oct 2026

Intro to Machine Learning — NYU College & Career Lab
~60 middle school students · Teachable Machine, ml5.js
Jul 2026

Soft Circuit Boards — ITP Camp, NYU
No-solder handmade circuits
Jun 2026

Teaching Assistant, Intro to Wearables — NYU ITP
Instructor: Daniel Ryan Johnston
Jan – May 2026

Teaching Assistant, Creative Computing — NYU IMA
Prof. Daniel Shiffman
Sep – Dec 2025

Research Assistant, Computational Text — NYU ITP
Prof. Daniel Shiffman · syllabus research on ML and accessibility
May – Aug 2025

PANELS & TALKS

Onassis ONX × American Academy in Rome, NYC
Panelist: They Make You Write Your Poems — Artistic Practices Amid Automated Cognition
Oct 2026

NYU Global Show & Tell
Lightning talk: "Open Tools, Soft Interfaces"
Oct 2025

FELLOWSHIPS & HONORS

Open Hardware Summit Fellow, OSHWA 2026

O'Reilly DEIJ Scholarship 2024

NYU Tisch Graduate Scholarship 2024

Bertelsmann Technology Scholarship, Udacity 2022

Grace Hopper Scholarship, AnitaB.org 2019 – 2022

Presidential Scholar, CUNY Lehman College 2020